Project Status Report

Introduction to Machine Learning

Project Topic

RNN/LSTM for Time Series Prediction — Stock Price Forecasting

Team Members

Ryan O'Neil (Rco220002), Section [YOUR SECTION] — individual project (team of one)

Technique / Algorithm to Implement

We will implement a Long Short-Term Memory (LSTM) recurrent neural network **from scratch** to perform time-series prediction on historical stock price data. The core algorithm — the LSTM cell (input, forget, and output gates and cell-state updates), the forward pass, backpropagation through time, and the gradient-descent training loop — will be coded by us using only NumPy, without any built-in deep-learning library. Pre-processing, data loading, and results evaluation will use standard libraries. Rather than training a separate model per stock, we will train a single "global" LSTM on windowed sequences pooled across multiple companies, allowing us to investigate whether pooling more companies improves the model's ability to learn general temporal patterns.

Dataset Details

Historical daily stock price data sourced from Kaggle (a single consistent multi-stock dataset so all companies share the same format) and hosted as CSV files on GitHub for reproducibility.

• Starting point: S&P 500 index (^GSPC), ~10 years of daily data ≈ 2,500 trading days.

• Scalable set: additional individual companies (e.g., AAPL, TSLA, and more) added incrementally to grow the training set.

• Features per record: Date, Open, High, Low, Close, Volume (Adjusted Close where available).

• Prediction target: daily closing price.

• Data distribution: non-stationary financial time series with varying volatility across companies (the index is relatively smooth while individual stocks such as TSLA are highly volatile). Each company's series is normalized (per-stock min-max scaling or conversion to returns) before pooling, so differing price levels do not dominate training.

Coding Language / Technique

Python. The LSTM is implemented from scratch using NumPy for all numerical operations. Supporting libraries (pre-processing and analysis only): pandas (loading/pre-processing), scikit-learn (train/test split and evaluation metrics), and matplotlib (visualization). No deep-learning frameworks (TensorFlow, PyTorch, Keras) are used to build or train the model.

Preliminary Results

Implementation in progress; no experimental results yet. Initial experiments will begin with the single-series S&P 500 model to validate the from-scratch LSTM (predicted-vs-actual price curve), followed by incremental addition of companies to the pooled training set, with each configuration recorded as a logged experiment.